IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies:

- SpaceX API calls and web scraping for data collection
- Data wrangling and transformation using pandas
- Exploratory Data Analysis (EDA) with Matplotlib, Seaborn, SQL
- Interactive visualizations using Folium and Plotly Dash
- Predictive modeling using classification techniques (SVM, RF, etc.)

Summary of all results:

- Random Forest Classifier achieved highest accuracy
- Launch site and payload strongly affect landing success
- Created interactive dashboard and Folium maps

Introduction

Project background and context:

- SpaceX aims to reduce the cost of space travel by reusing the first stage of Falcon 9 rockets. Determining the likelihood of a successful landing supports mission planning.

Problems you want to find answers:

- What features influence landing success?
- Can we predict successful landings?
- What spatial factors (e.g., proximity to coastline) matter?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
    - SpaceX API used for mission records (https://api.spacexdata.com/v4/rockets/)
    - Web scraping for payload, orbit, and booster data (https://en.Wikipedia.org/wiki/List_of_Falcon/_9/_and_Falcon_Heavy_launches)
- Perform data wrangling
    - Removed irrelevant columns, filled missing values.
    - Converted categorical to numerical features via One-Hot Encoding.
    - Normalized features.
- Perform exploratory data analysis (EDA) using visualization and SQL

Methodology

Executive Summary

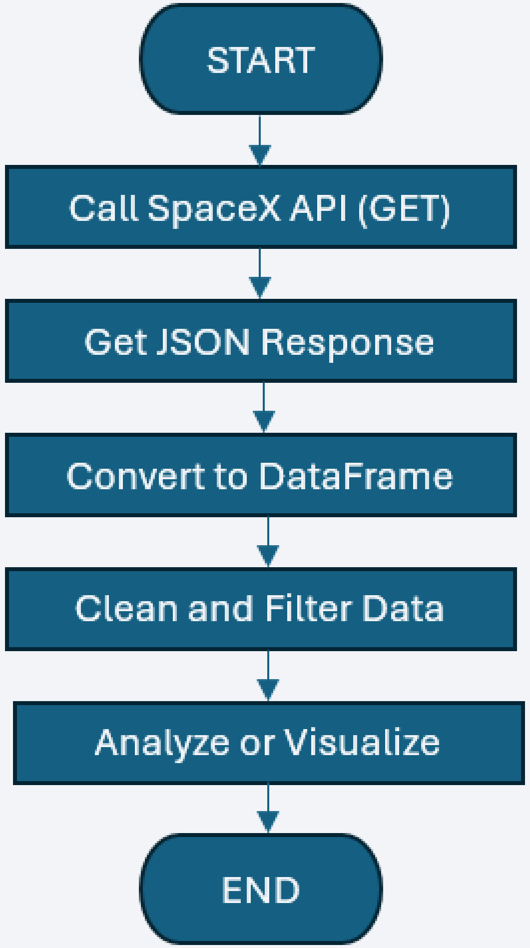
- Perform exploratory data analysis (EDA) using visualization and SQL
  - Visualization via Matplotlib, Seaborn.
  - SQL queries used for structured analysis.
- Perform interactive visual analytics using Folium and Plotly Dash
  - Folium for geographical map analysis.
  - Plotly Dash for launch outcome trends.
- Perform predictive analysis using classification models
  - Built models: Logistic Regression, SVM, Decision Tree, KNN, Random Forest.
  - Used GridSearch for hyperparameter tuning.
  - Evaluated models via accuracy, F1 score, confusion matrix.

Data Collection

- Retried launch data via GET request to the SpaceX API
- Parsed JSON response with .json() and converted it to a pandas DataFrame using json_normalize()
- Performed data cleaning, checked missing values, and filled in missing entries as needed
- Scaped Falcon 9 launch records from Wikipedia
- Extracted and parsed HTML tables
- Converted tables into a pandas DataFrame for analysis

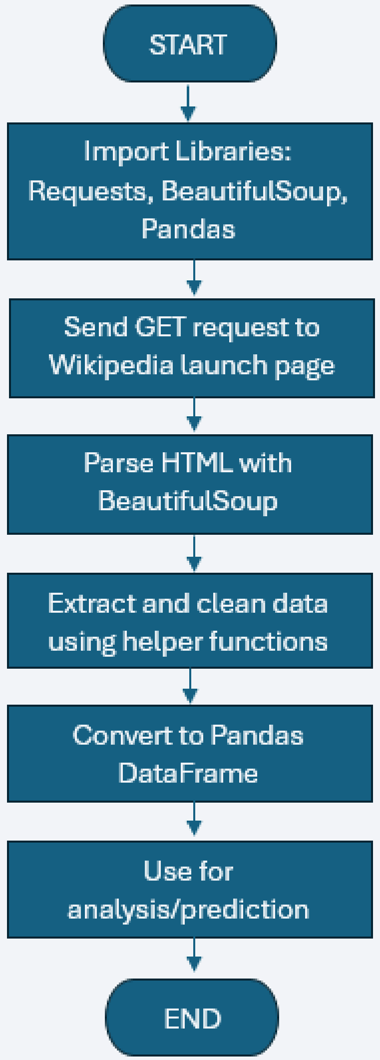
Data Collection – SpaceX API

- Used the SpaceX API to retrieve mission launch data.
- The link to the notebook is: https://github.com/jmvfernandez/data-science-notebook/blob/main/1.%20SpaceX%20Data%20Collection%20API.ipynb



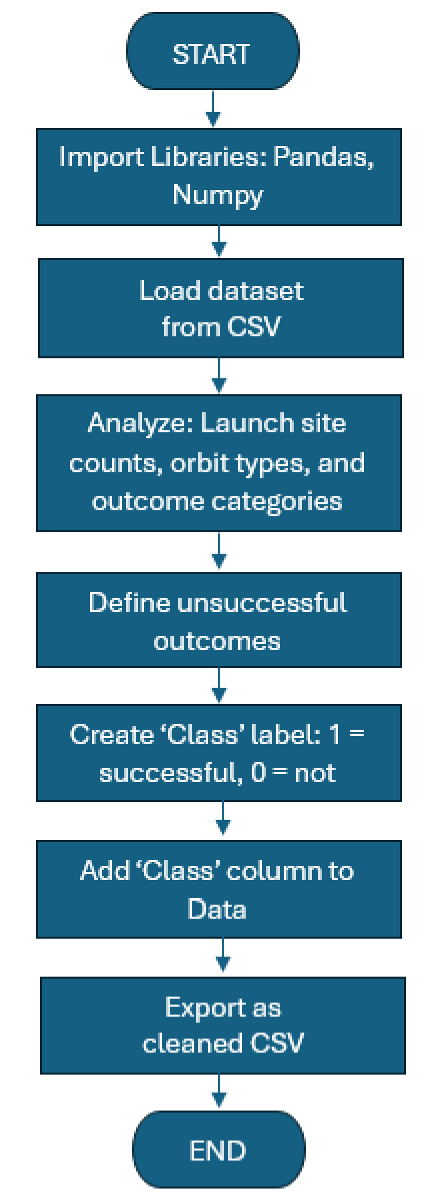
Data Collection - Scraping

- SpaceX launch data can be sourced from Wikipedia.
- Data is extracted following a defined flowchart and then stored for further analysis.
- The link to the notebook is: https://github.com/jmvfernandez/data-science-notebook/blob/main/2.%20Webscraping.ipynb



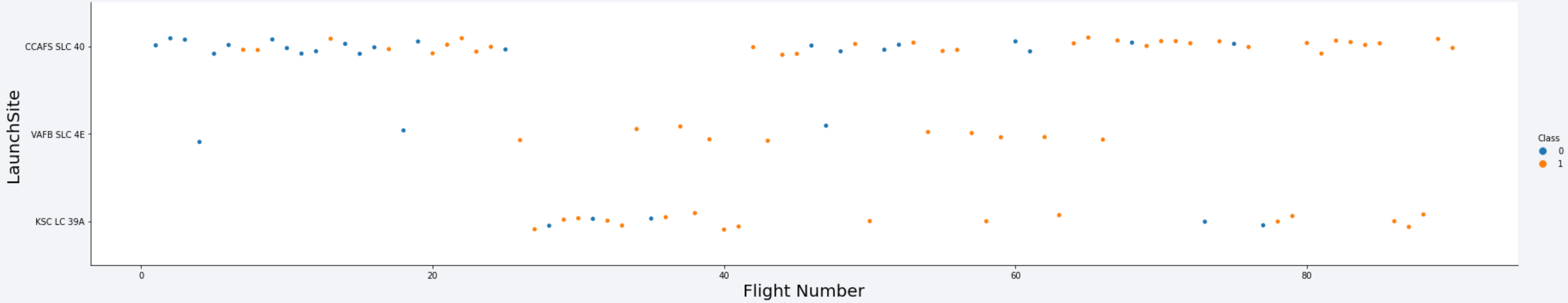
Data Wrangling

- Loaded SpaceX Falcon 9 launch dataset.
- Checked for and handled missing values.
- Identified numerical and categorical columns.
- Analyzed launch sites, orbit types, and landing outcomes.
- Grouped outcomes into success or failure using a new Class label.
- Added Class column to the dataset for supervised learning.
- Exported the cleaned dataset for further analysis and modeling.
- The link to the notebook is: https://github.com/jmvfernandez/data-science-notebook/blob/main/3.%20Data%20Wrangling.ipynb



EDA with Data Visualization

- Created box plots, scatter plots, and bar charts.
- Identified trends in payload orbit, and site success rates.



- GitHub URL: https://github.com/jmvfernandez/data-science-notebook/blob/main/4.%20EDA%20with%20Data%20Visualization.ipynb

EDA with SQL

- Retrieved all unique SpaceX launch site names.
- Identified top 5 launch sites starting with ‘CCA’.
- Calculated total payload mass for NASA (CRS) missions.
- Computed average payload mass for booster version F9 v1.1.
- Found the date of the first successful ground pad landing
- Listed boosters with successful drone ship landings and payloads between 4000-6000 kg.
- Counted total successful and failed mission outcomes.

EDA with SQL (continuation)

- Identified booster versions that carried the maximum payload.
- Extracted failed drone ship landings in 2015 with booster and site information.
- Ranked landing outcomes (e.g., Success, Failure) between 2010-06-04 and 2017-03-20.
- GitHub URL: https://github.com/jmvfernandez/data-science-notebook/blob/main/5.%20EDA%20with%20SQL.ipynb

Build an Interactive Map with Folium

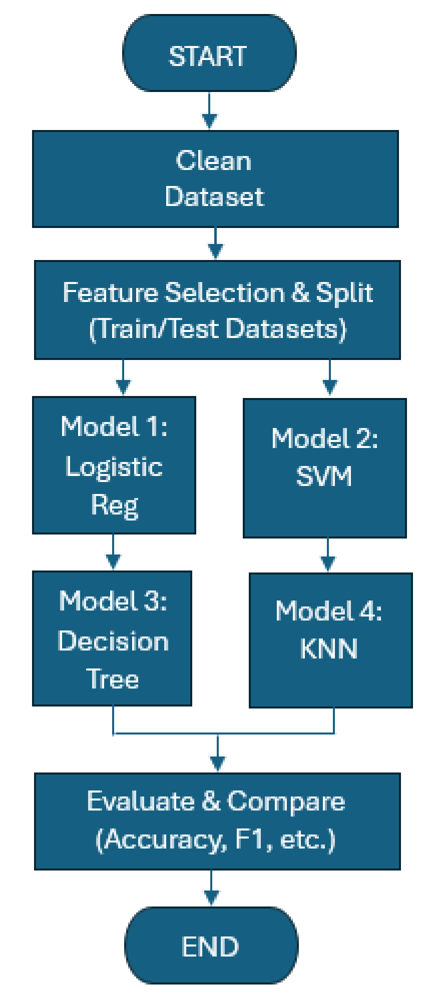
- Add popups or tooltips to markers to display additional information (e.g., site name, launch date)
- Use different colors or icons for different outcomes (e.g., success/failure)
- Include a legend to explain the map symbols.
- Overlay heatmaps to show launch density or success rates by location.
- Enable interactive filtering (e.g., by booster version, year) to make the map more exploratory
- Consider using GeoJSON layers for better accuracy (e.g., state borders or coastlines)
- GitHub URL: https://github.com/jmvfernandez/data-science-notebook/blob/main/6.%20Interactive%20Map%20with%20Folium.ipynb

Build a Dashboard with Plotly Dash

- Add a dropdown menu to filter by booster version, launch outcome, or year.
- Include a bar chart to show success rates per launch site.
- Add a range slider to dynamically filter payload mass.
- Use callbacks to make plots reactive to user input, improve interactivity.
- Display hover tooltips with more detailed info (e.g., payload name, mission).
- Consider adding a map integration to complement the dashboard with spatial context.
- GitHub URL: https://github.com/jmvfernandez/data-science-notebook/blob/main/7.%20SpaceX%20Dash%20App.py

Predictive Analysis (Classification)

- Include a confusion matrix for each model to visualize true vs. predictive values.
- Present a performance comparison table for better clarity
- Perform cross-validation to ensure model robustness.
- Use feature importance (from Decision Tree or Logistic Regression) to identify key predictors).
- Consider hyperparameter tuning using GridSearchCV for optimal results.
- Include ROC curves to visualize trade-off between sensitivity and specificity
- GitHub URL: https://github.com/jmvfernandez/data-science-notebook/blob/main/8.%20SpaceX%20Machine%20Learning%20Prediction.ipynb



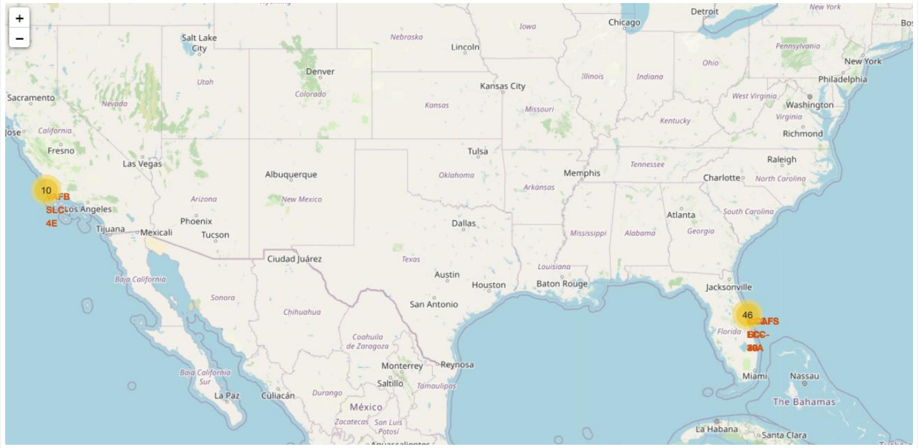
Results

Exploratory Data Analysis (EDA) Results:

- SpaceX has utilized four distinct launch sites for its missions.
- Initial launches were conducted for both SpaceX's internal projects and NSA collaborations.
- The average payload carried by the Falcon 9 v1.1 booster is approx. 2,928kg.
- The first successful landing occurred in 2015, around five years after the company's initial launch.
- Multiple Falcon 9 booster versions successfully landed on drone ships, often when carrying payloads above the average.
- Mission success rates approached nearly 100%, reflecting SpaceX's rapid learning curve and engineering improvments

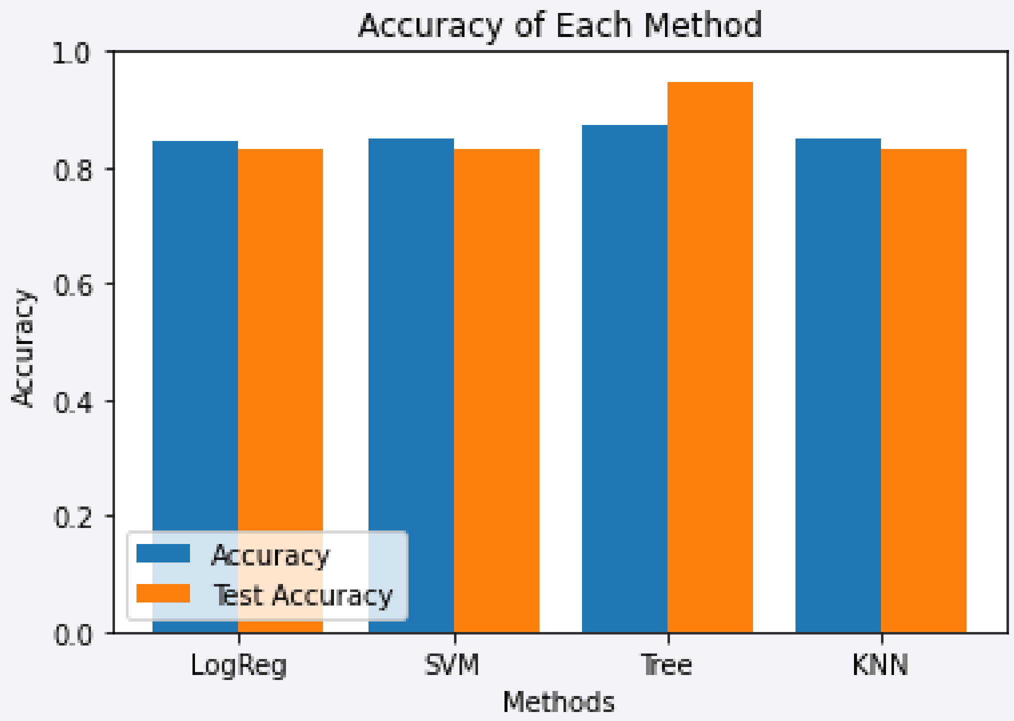
Results (continuation)

- Mission success rates approached nearly 100%, reflecting SpaceX's rapid learning curve and engineering improvements.
- In 2015, two booster versions (F9 v1.1 B1012 and F9 v1.1 B1015) failed to land on drone ships.
- Landing outcomes have consistently improved over the years, showing a trend toward increased reusability and reliability.
- Launch sites are usually in safe, coastal areas with good logistics.
- Most launches take place on the East Coast.



Results (continuation)

- Predictive analysis revealed that the Decision Tree Classifier is the most effective model for predicting successful landings, achieving over 87% training accuracy and more than 94% accuracy on test data.

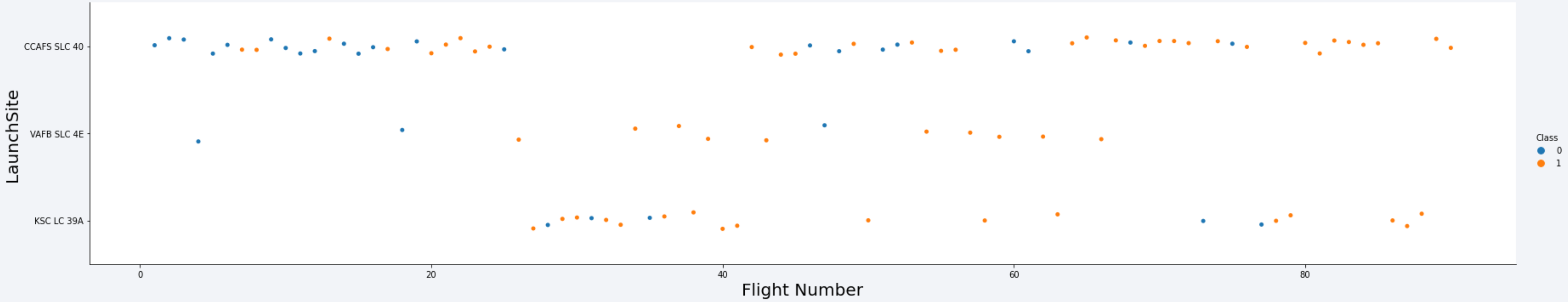


Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

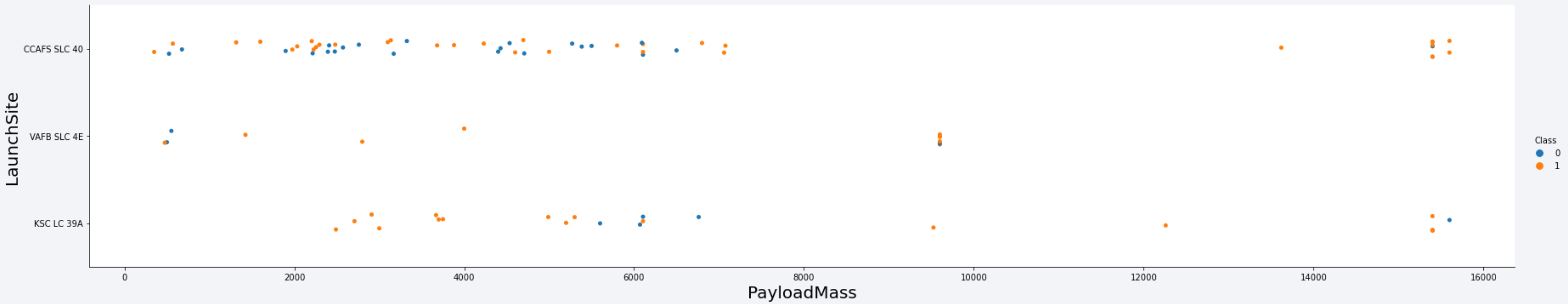
- The plot indicates that CCAFS SLC-40 is currently the most successful launch site, with the highest number of recent successful launches.
- VAFB SLV-4E and KSC LC-39A follow in second and third place, respectively.
- Overall, the success rate has steadily improved over time, reflecting advancements in launch technology and operations.



Payload vs. Launch Site

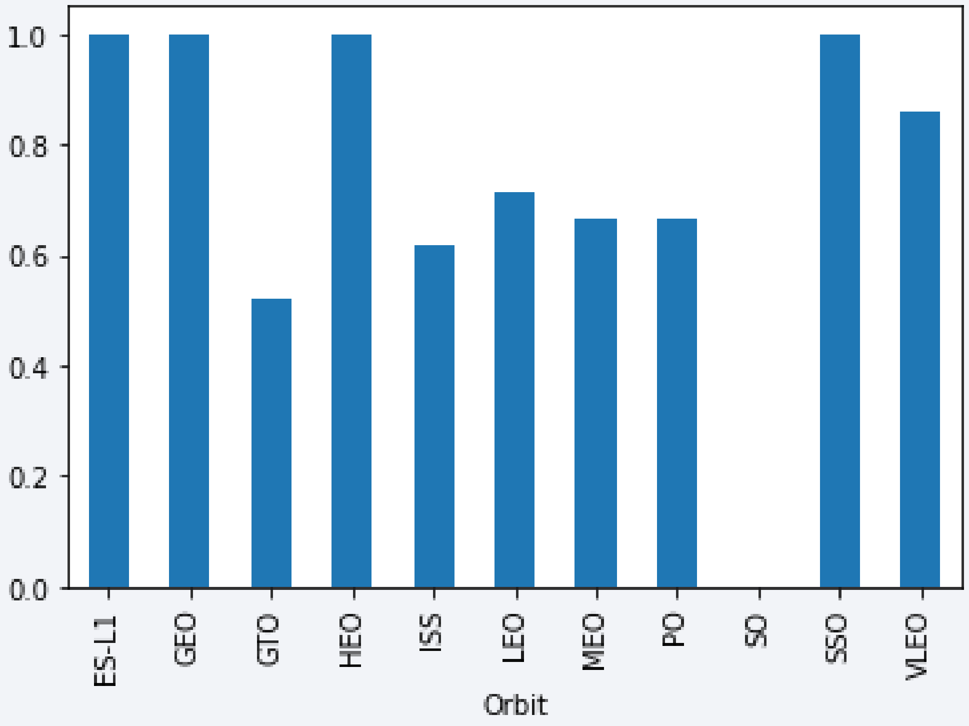
- Payloads exceeding 9,000kg. Roughly the weight of a school bus show an excellent success rate in launch and landing outcomes.

- Payloads above 12,000kg. Appear to be launched exclusively from CCAFS SLC-40 and KSC LC-39A, likely due to their superior infrastructure and launch capabilities.



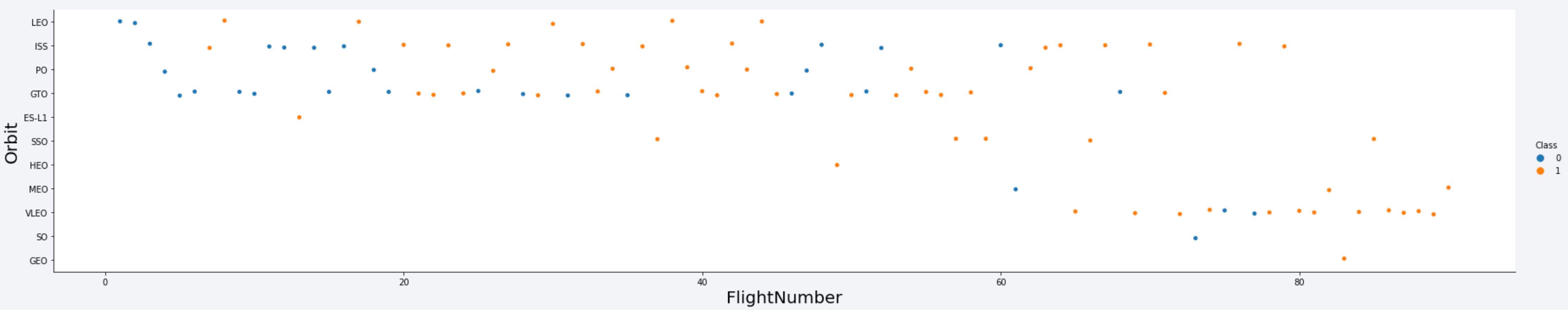
Success Rate vs. Orbit Type

- The highest success rates are observed in mission targeting the ES-L1, GEO, HEO, and SSO orbits.
- These are followed by missions to VLEO (above 80% success rate) and LEO (above 70% success rate).



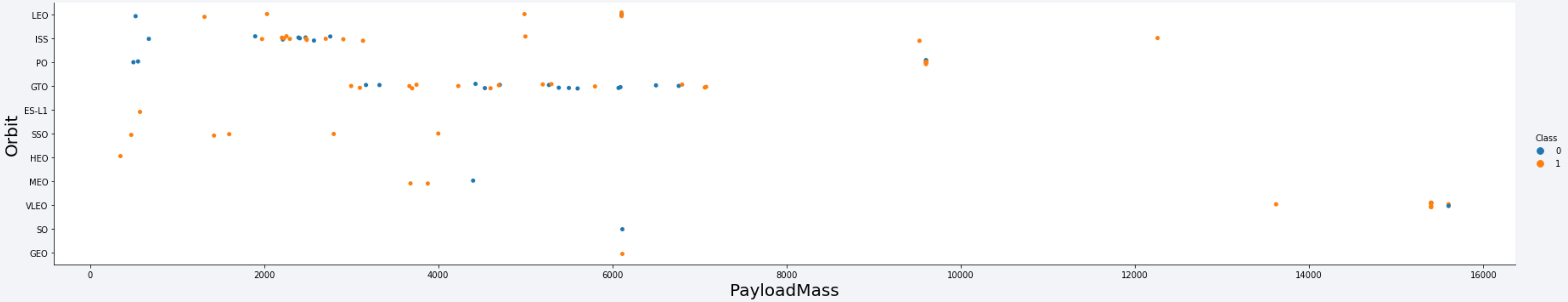
Flight Number vs. Orbit Type

- Success rates have consistently improved across all orbit types over time.
- The recent rise in VLEO missions suggests it may be an emerging business opportunity.



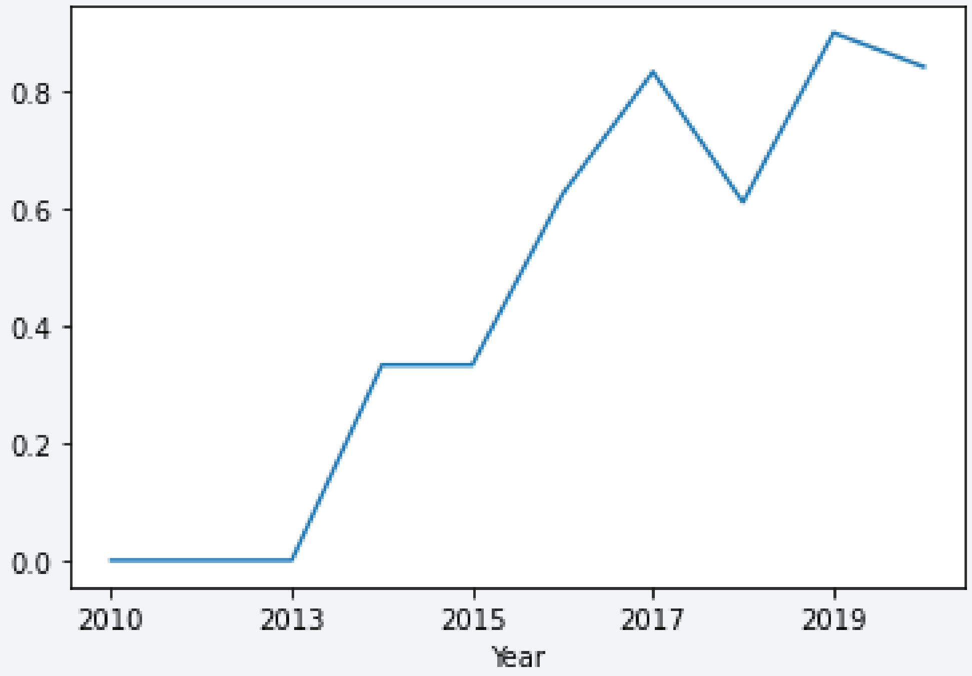
Payload vs. Orbit Type

- There appears to be no clear correlation between payload mass and success rate for GTO (Geostationary Transfer Orbit) missions.
- ISS (International Space Station) orbit supports the widest range of payloads and maintains a high success rate.
- Very few launches target these orbits compared to others.



Launch Success Yearly Trend

- The success rate began to rise steadily starting 2013 and continued to improve through 2020.
- The initial three years (2010-2012) appear to have been phase of adjustments, testing, and technological refinement.



All Launch Site Names

According to the dataset, there are four distinct launch sites used by SpaceX:

- CCAFS LC-40
- CCAFS SLC-40
- KSC LC-39A
- VAFB SLC-4E

These launch sites were identified by extracting the unique values from the launch_site column in the dataset.

Launch Site Names Begin with 'CCA'

- Five records where the launch site name begins with 'CCA':

Date	Time UTC	Booster Version	Launch Site	Payload	Payload Mass kg	Orbit	Customer	Mission Outcome	Landing Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attemp

- These entries represent sample launches conducted at Cape Canaveral, showcasing a subset of the missions launched from this region.

Total Payload Mass

Total Payload Carried by Boosters for NASA Missions:

The total payload is **111.268kg.** Calculated by summing all payloads with mission codes containing ‘CRS’, which are associated with NASA’s Commercial Resupply Services.

Total Payload (kg)
111.268

Average Payload Mass by F9 v1.1

The average payload mass is **2.928kg.**, calculated by filtering the dataset for the F9 v1.1 booster version and computing the mean payload value.

Avg Payload (kg)
2.928

First Successful Ground Landing Date

The Successful Ground Pad Landing:

Min Date
2015-12-22

The first successful booster landing on a ground pad occurred on December 22, 2015. This was identified by filtering the dataset for successful landings on ground pads and retrieving the earliest launch date.

Successful Drone Ship Landing with Payload between 4000 and 6000

The following booster versions successfully landed on drone ships while carrying payloads between 4,000kg. And 6,000kg.:

Booster Version
F9 FT B1021.2
F9 FT B1031.2
F9 FT B1022
F9 FT B1026

These results were obtained by applying filters on landing success, landing type (drone ship), and payload mass range, followed by selecting distinct booster versions.

Total Number of Successful and Failure Mission Outcomes

Mission Outcome Summary:

Mission Outcome	Occurrences
Success	99
Success (payload status unclear)	1
Failure (in flight)	1

By grouping the dataset based on mission outcomes and counting the number of records in each category, we generated a summary of **successful vs. failed launches.**

Boosters Carried Maximum Payload

Boosters that carried the maximum payload mass:

Booster Version (...)
F9 B5 B1048.4
F9 B5 B1048.5
F9 B5 B1049.4
F9 B5 B1049.5
F9 B5 B1049.7
F9 B5 B1051.3

Booster Version
F9 B5 B1051.4
F9 B5 B1051.6
F9 B5 B1056.4
F9 B5 B1058.3
F9 B5 B1060.2
F9 B5 B1060.3

The following boosters carried the maximum payload mass recorded in the dataset, demonstrating their high capacity and performance.

2015 Launch Records

Failed Drone Ship Landings in 2015:

Booster Version	Launch Site
F9 v1.1 B1012	CCAFS LC-40
F9 v1.1 B1015	CCAFS LC-40

The table above lists the only two failed landing outcomes on drone ships in the year 2015, including their booster versions and launch site names.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

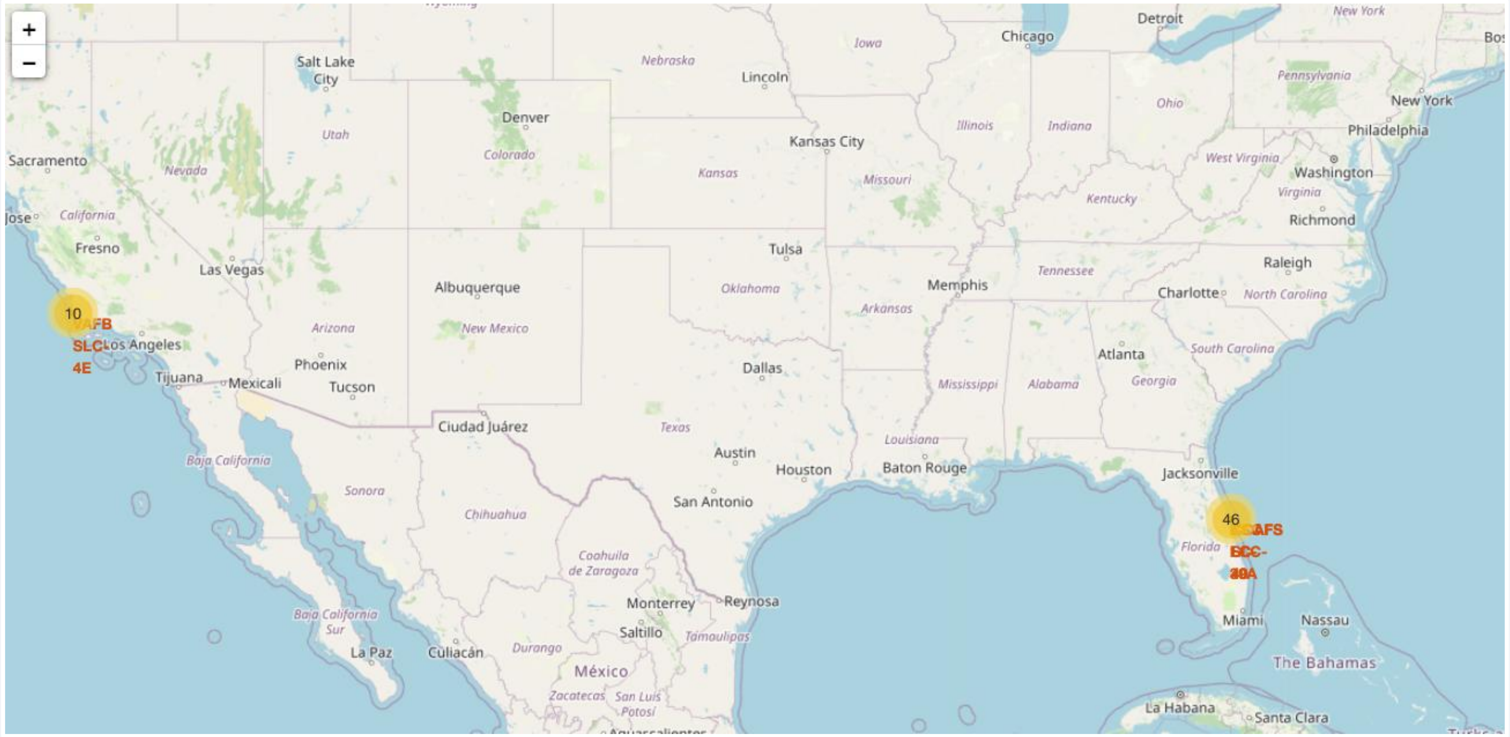
This analysis highlights the importance including “No attempt” outcomes in the evaluation, as they significantly impact the overall landing success rate.

Landing Outcome	Occurrences
No attempt	10
Failure (drone ship)	5
Success (drone ship)	5
Controlled (ocean)	3
Success (ground pad)	3
Failure (parachute)	2
Uncontrolled (ocean)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

All Launch Sites



Launches sites are typically located near the sea, likely for safety reasons, but they also remain accessible via nearby roads and railways to ensure efficient logistics.

Launch Outcomes by Site

Example of launch outcomes at KSC LC-39A:

Green markers represent successful launches, while red markers denote failures.

Logistics & Safety

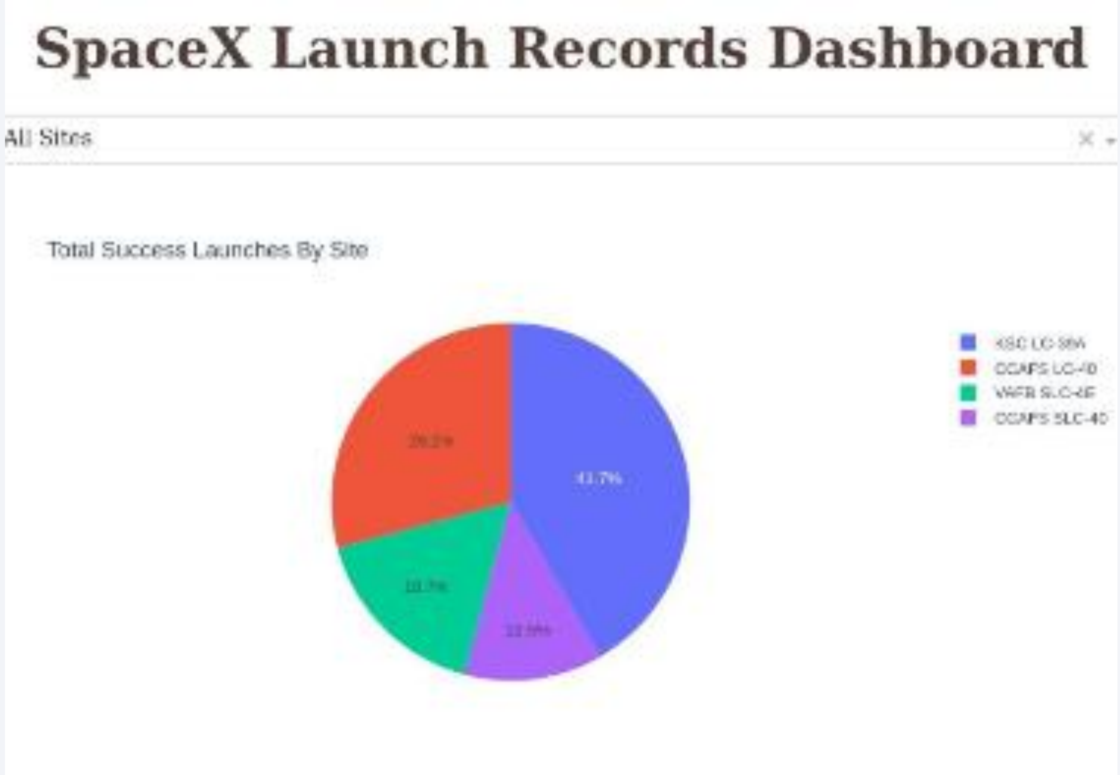
The launch site KSC LC-39A offers favorable logistical conditions, situated close to major roads and railways while maintaining a safe distance from densely populated areas.

Section 4

Build a Dashboard with Plotly Dash

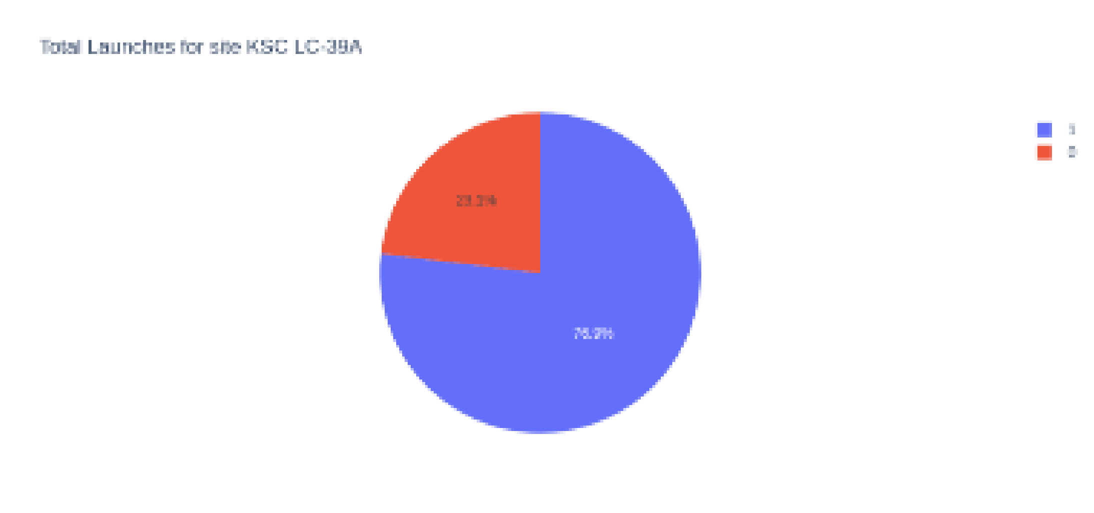
Successful Launches by Site

The success of space missions is strongly influenced by the launch site's location. Factors like proximity to the ocean, low nearby population, good logistics, and favorable conditions contribute to higher success rates. Sites like KSC LC-39A and CCAFS SLC-40 demonstrate this advantage.



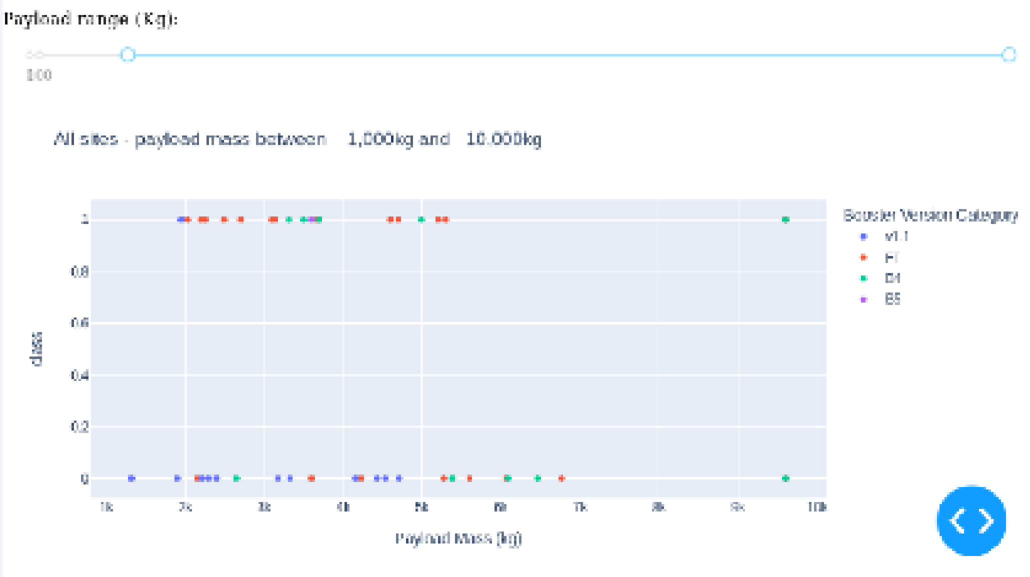
Launch Success Ratio for KSC LC-39A

At KSC LC-39A, 76.9% of launches have been successful, indicating a strong operational performance over the years.



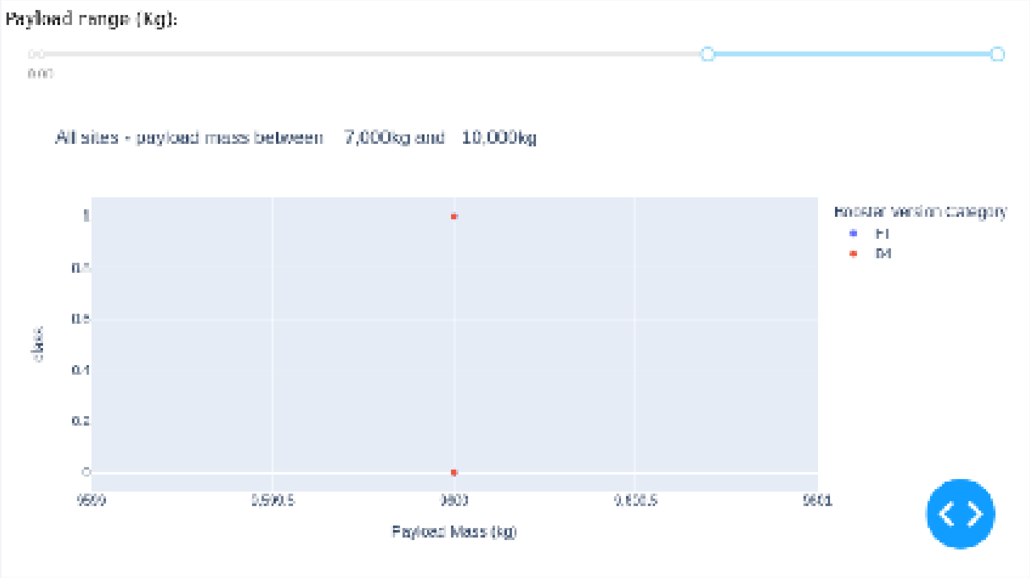
Payload vs. Launch Outcome

Analysis shows that missions carrying payloads under 6,000 kg and using Falcon 9 FT boosters achieved the highest success rates among all configurations, making them a reliable choice for mid-weight payload deliveries.



Payload vs. Launch Outcome (continuation)

Due to the limited number of launches exceeding 7,000 kg in the dataset, it is difficult to draw meaningful conclusions about their risk profile. More data is needed to determine whether higher payloads impact mission success rates.

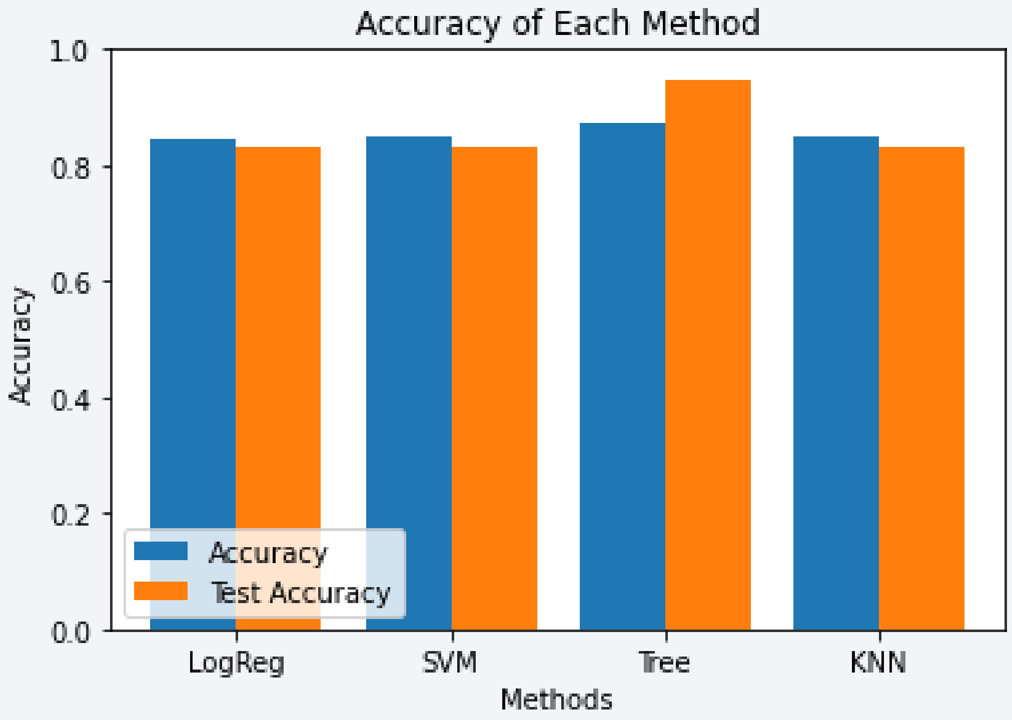


Section 5

Predictive Analysis (Classification)

Classification Accuracy

Among the four models tested, the Decision Tree Classifier outperformed the others with an accuracy of over 87%. This suggests it is the most effective model for predicting mission outcomes based on the available features.



Confusion Matrix

The high values in the diagonal of the confusion matrix highlight the classifier's effectiveness in correctly identifying both successful and failed launches, reinforcing its overall accuracy.

Conclusions

Multiple data sources were analyzed, allowing for increasingly refined conclusions throughout the process. The findings include:

- KSC LC-39A emerged as the most reliable and strategically located launch site.
- **Payloads exceeding 7,000 kg** are associated with lower risk, based on available data.
- While most mission outcomes are successful, **landing success rates have shown a clear upward trend over time**, likely due to advancements in rocket design and operational processes.
- The **Decision Tree Classifier** demonstrated high accuracy and can effectively predict successful landings, supporting informed decision-making and potentially enhancing profitability.

Appendix

- GitHub: https://github.com/jmvfernandez/data-science-notebook.git
- SQL Queries
- Folium & Dash Screenshots
- Model Evaluation Charts

Thank you!